MM7025 - COMPUTER ALGEBRA, COMPUTER EXERCISES 2

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This time it is most convenient to hand in a .tex-file where you mix code and comments. Work in the same constellation as before. If you have questions, need more hints, or if something is unclear, let me know.

(1) Redo 1.4.3b and 1.4.3c from Exercise 1 in the Textbook exercises, but this time using M2. There are several ways to do this exercise in M2. I want you to do it
   - by using "==" directly applied on the ideals,
   - by computing Gröbner bases for the ideals and checking that they are equal by looking at the output,
   - by checking ideal membership using the "%"-operator (in two directions, first check that every generator on the left hand side is an element of the ideal on the right hand side, and then do the opposite).

(2) Let $f_1 = x^2+y+z+1$, $f_2 = x^2+y^2+3x$, $f_3 = y^2+xz+x+1$, $f_4 = y^2+y+x+5$ be elements in $\mathbb{C}[x, y, z]$.

   Show that there is no common solution to the system of equations $f_1 = f_2 = f_3 = f_4 = 0$, that is, show that $V(\langle f_1, f_2, f_3, f_4\rangle) = \{\}$. Hint: Use that a Gröbner basis of an ideal is a generating set of the ideal.

(3) Find four solutions to the following system of equations over $\mathbb{C}$:

$$\begin{cases} 5z^2 - 6x + 3y + 10z - 12 = 0 \\ 5yz - 6x + 18y - 5z - 12 = 0 \\ 5xz - 3x + 9y - 5z - 6 = 0 \\ 5y^2 - 2x + y - 4 = 0 \\ 5xy - x - 2y - 2 = 0 \\ 5x^2 + 7x - 6y - 6 = 0 \end{cases}$$

   Hint: Compute a GB with respect to Lex for the polynomials on the left hand side and then use extension.

(4) We have seen that different monomial orderings give rise to different $\mathrm{LM}(I)$s. For instance, if you run the code

```
R=QQ[x,y,z, MonomialOrder =>Lex]
I = ideal(x^2+x+z,y^2+x+y^3+1,z^2+x^2+x)
flatten entries gens gb I
```

   you see that with respect to Lex and $x > y > z$, we get $\mathrm{LM}(I) = \langle z^2, y^6, x\rangle$ (we can write $\mathrm{LM}_{lex:x>y>z}(I) = \langle z^2, y^6, x\rangle$ to be more specific), while if you run the code

```
R=QQ[x,y,z, MonomialOrder =>GRevLex]
I = ideal(x^2+x+z,y^2+x+y^3+1,z^2+x^2+x)
flatten entries gens gb I
```

you see that with respect to Graded Reverse Lex and $x > y > z$, we get $\mathrm{LM}(I) = \langle z^2, y^3, x^2 \rangle$.

Play around with some other orderings (it is easiest to fix Lex and interchange the ordering between x, y, z) to find at least two other $\mathrm{LM}(I)$s, so that you in total have at least four $\mathrm{LM}(I)$s.

For each of the four $\mathrm{LM}(I)$, list the monomials which are not in $\mathrm{LM}(I)$ and count them.

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